

Crime and What Else? Investigating the Predictive Power of Social Factors on Incarceration Rates

Emery Kerani

DATA 510 Data Science Capstone · Summer 2026



Abstract

Crime is often taken for granted as the primary driver of incarceration in the United States, without accounting for underlying factors (e.g. poverty) or large-scale policies that expand the prison-industrial complex. This project aims to evaluate the truth of this dominant narrative by utilizing machine learning to assess how well a variety of factors are able to predict incarceration rates. Among both single-feature models and a model using all features to generate predictions, police spending had the lowest error rate and highest feature importance, indicating that government spending on police is more closely associated with incarceration rates than crime.



Introduction & Motivation

Problem & question: Rather than simply accepting the US's continued reliance on incarceration and its associated harms, we could all benefit from considering how the criminal legal system actually works and whose interests it protects. To that end, **this project seeks to interrogate whether incarceration is better predicted by crime or by social factors (such as race, poverty, or government spending on police).**

Impact & stakeholders: The key stakeholders in this work are those who are most often targeted by criminalization and incarceration, primarily poor and BIPOC folks. Additionally, communities across the country are continually harmed by increasing incarceration rates, as incarceration has been shown to negatively impact both community health and poverty [1]. By critically examining the circumstances that can contribute to incarceration, this project seeks to uplift the work of organizations such as Transformative Justice Community and the Oregon Justice Resource Center, which advocate for solutions to harm that shift our paradigm away from imprisonment.



Background & Related Work

For decades, data science and statistics have played a massively important role in prison reform and abolition causes; as just one example, we have known for well over three decades that the population of Black prisoners is disproportionate to the Black population in the US [1]. Data concerning overall prison populations have also been key in tracking the effects of legislation concerning crime and sentencing, many of which contributed to astronomical growth of the US prison population [2]. Police spending is another facet of prior research, particularly when it comes to questions of how to effectively and durably address crime—while increased policing may reduce crime in the short term, studies suggest that diverting money from policing into social services more reliably reduces crime in the long term [3].

To this point, most of the work applying machine learning algorithms to data concerning the prison system has focused on predicting or classifying individual behaviors or risk factors. This project takes a more abstracted view of the legal system and its contributing factors as a whole.

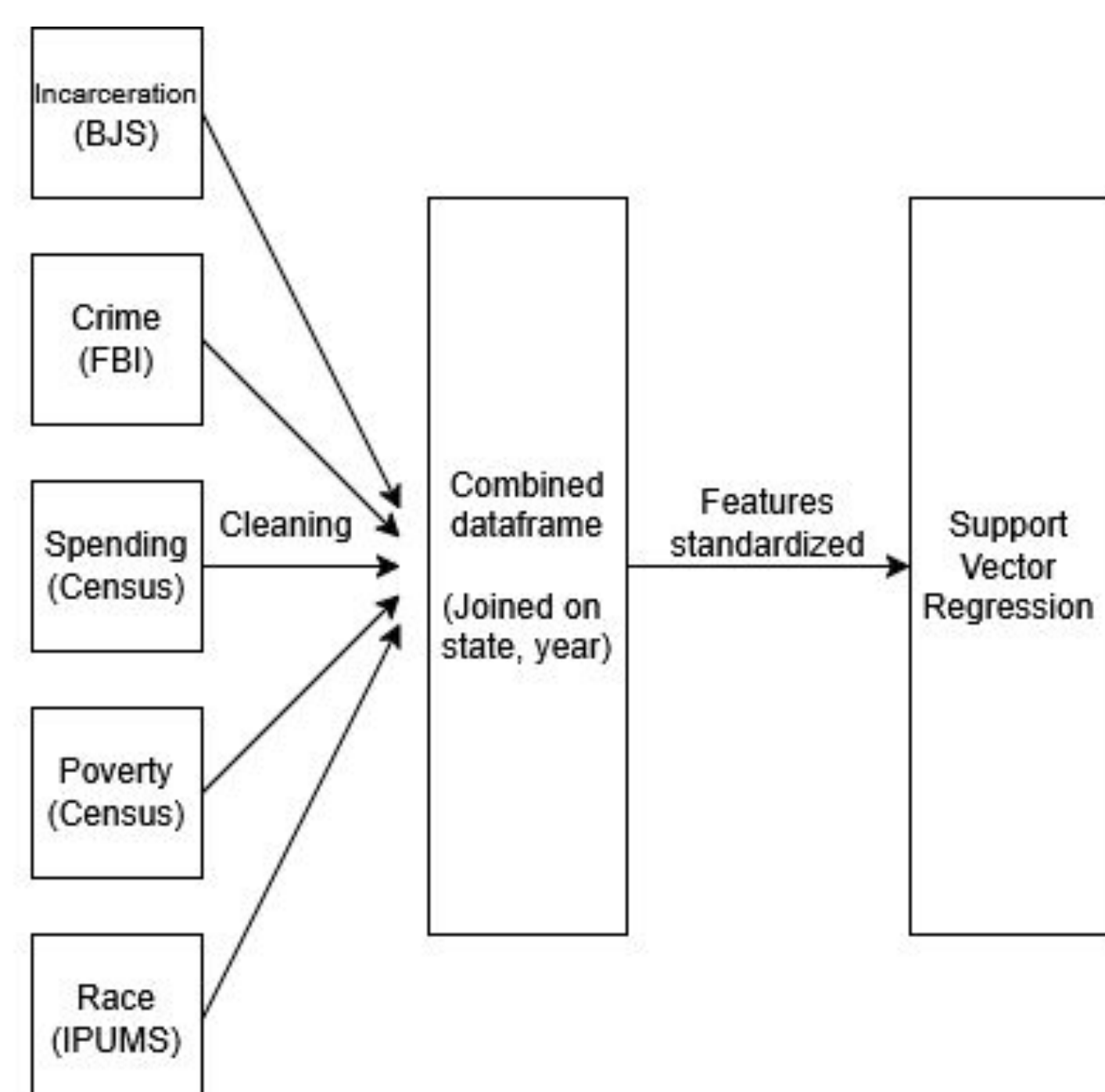


Data, Engineering & Ethics

Sources & scale: The final data set for this project was constructed by combining data from 5 disparate sources: The Bureau of Justice's Corrections Statistical Analysis Tool, the FBI's Summary Reporting System, government spending data from the US Census via the Urban Institute, historical poverty statistics from the Census, and race data from IPUMS' Current Population Survey. These data span 40 years, from 1980-2020, and all 50 states are represented each year, for a total of 2,050 rows.

Pipeline: Each data source was ingested as a one-time, manual download, or in the case of IPUMS loaded directly into R. Once ingested, the format of each dataset was standardized and the sources were combined into one frame based on state-year pairings.

Ethics & responsible use: All of the data used for this project were already highly aggregated, with the exception of the IPUMS data—which were available at the household level with identifying characteristics already removed, and then were manually aggregated using provided weights. They thus pose little (if any) privacy risks to individuals, who had to have consented to be included in the original surveys. Importantly, this is a correlational study, and as such no claims should be made about causation. Sources are cited according to the requirements laid out by each publishing body.



Methods & Evaluation

Approach: This project uses support vector regression (SVR) as the primary method of achieving the goals laid out on this poster. SVR was chosen because of its ability to deal with complex, non-linear data while remaining more straightforward than methods such as XGBoost; the relatively small size of the combined dataset also makes SVR a good fit as it optimized to work with smaller datasets. All models were created using SVR's default parameters to ensure fairness when comparing results between predictor variables. These models were trained on individual predictor variables across time, using the national incarceration rate as the label for each, in order to assess the predictive power of each variable in isolation. The variables were then combined to train a final model, which was used to find the feature importance of each variable.

Validation: During the data cleaning process all variables were converted to rates (per 100k people), and before training any models all predictive features were standardized using a robust scaler to ensure no variable had undue influence on the model due to size or scale. Additionally, the correlation coefficient was used as a baseline metric to assess the linear relationship between all variables. After training models, the r^2 score and root mean squared error was calculated for each to test the capability of each variable to explain the national incarceration rate and the average error in each model's predictions, respectively. There was no cutoff for these measures, as the goal was simply to compare across models.



Results & Visualizations

As expected, none of the features were linearly correlated with state incarceration rates; all had a “weak” correlation coefficient of less than 0.5. When using single features (i.e. only police spending) as predictors of the national incarceration rate, the models generated performed poorly: only two features (police spending at 89.01 and property crime rate at 94.41) had RMSEs of less than 100 people per 100k; additionally, the r^2 scores (none above 0.3) indicate that no individual feature was able to explain the variance in the national incarceration rate particularly well. However, out of these poor models, the one trained on police spending rates had the best RMSE and r^2 scores. A model trained on all features performed markedly better, with an RMSE of 53.65 and an r^2 score of 0.74. Relative to all other features, police spending rate was the most most important in generating this model's predictions with a permutation importance of 0.16, which shows that the model's performance suffered the most when police spending data was shuffled.

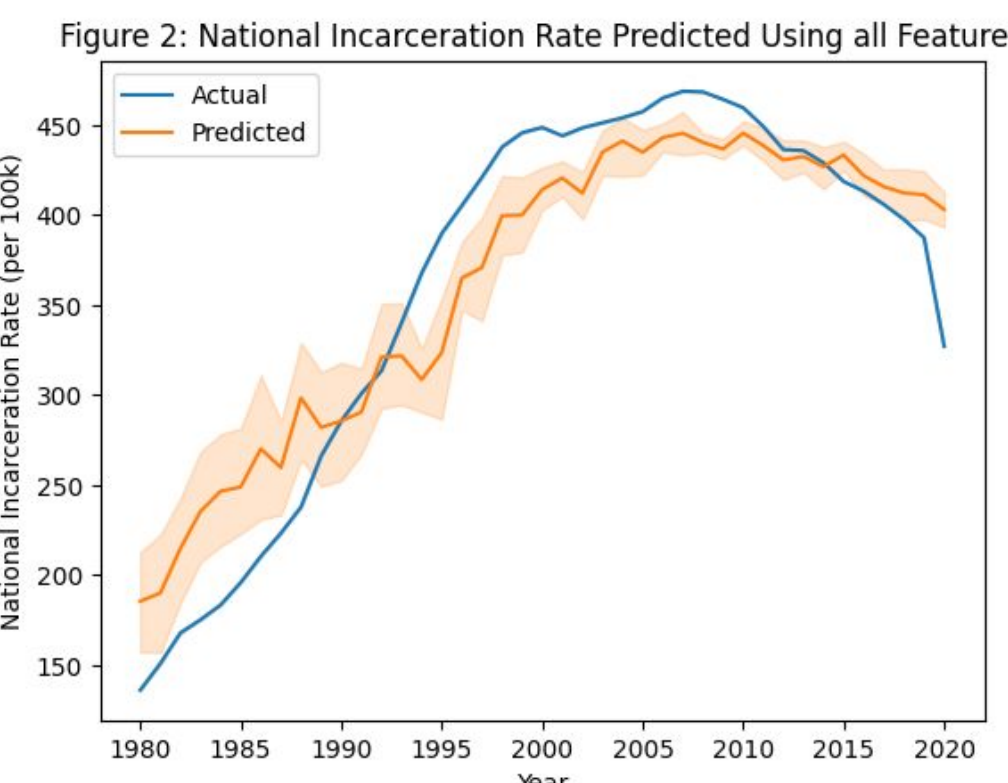
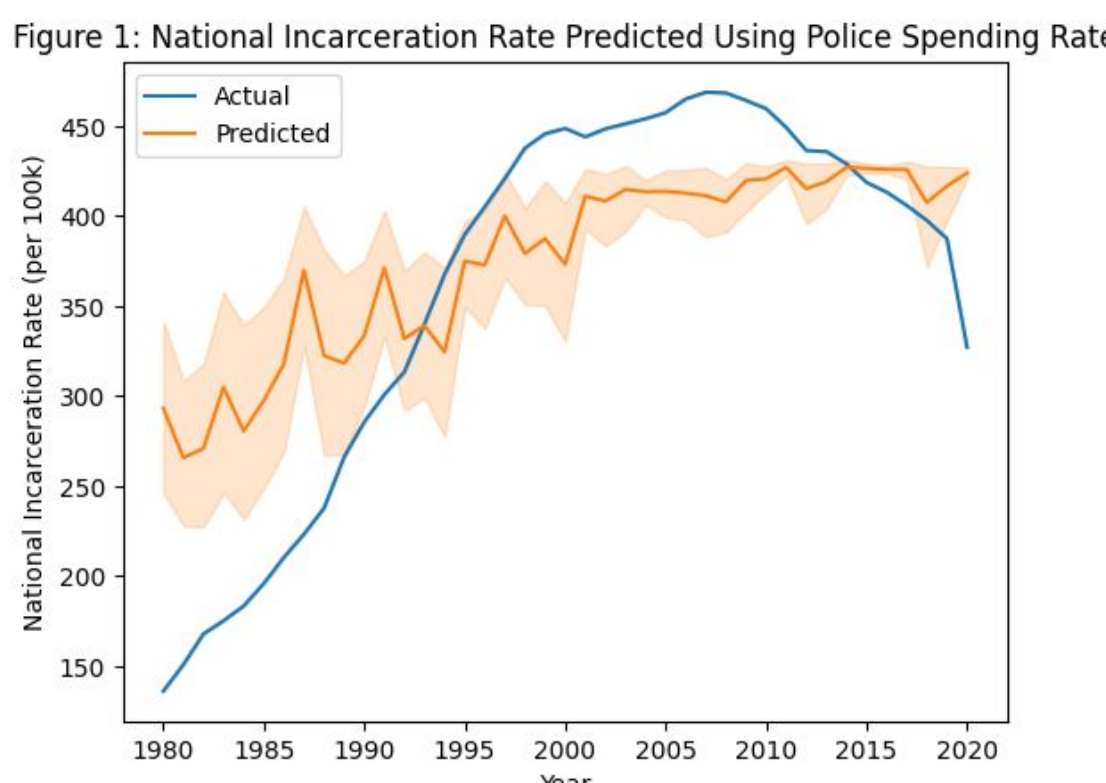


Figure 3: All Model Results

	Correlation	National RMSEs	National r^2 scores	National feature importance	State RMSEs	State r^2 scores	State feature importance
Violent crime rate	0.31	113.37	-0.14	0.03	160.21	0.20	0.19
Property crime rate	-0.09	94.41	0.21	0.12	175.91	0.04	0.13
Police spending rate	0.42	89.01	0.30	0.16	156.73	0.23	0.27
Poverty rate	0.15	113.83	-0.15	0.03	177.62	0.02	0.09
Non white rate	0.45	107.26	-0.02	0.03	140.09	0.39	0.27



Discussion, Limitations & Conclusions

Discussion: The results above indicate that across the board, police spending rate serves as the best (or most important) predictor for national incarceration rate using SVR models. In all evaluation metrics, property crime rates were a close second, meaning that where crime was associated with incarceration rates in this project it was primarily property crime and *not* violent crime. Additionally, the performance difference between single-feature and combined-feature models confirms that none of these variables truly exist in isolation, and all must be accounted for in order to build a fuller picture of what might influence incarceration in the United States.

Limitations & future work: As this project only explored associations between variables and not causality, no claims can be made about how any of the variables used might *cause* incarceration. It must also be noted that these data and the features used are by nature interconnected; some of the features used to generate may have impacted others (such as police spending and crime rates, which were based on reports/arrests made). Future work could explore variations in the type of incarceration (e.g. jails vs. prisons, men vs. women) or concentrate on smaller geographical areas such as counties to incorporate more data.

Conclusions: Police spending served as the most influential predictor of national incarceration rates. Any efforts to reduce incarceration should keep police and legal system spending in mind as a factor when engineering reforms.

Reproducibility & References

Repository: [<https://github.com/eskerani/prison-predictor-capstone>] · Key dependencies: [Python 3.11, pandas, scikit-learn, seaborn, numpy, matplotlib, dataframe_image; R 4.5.2, tidyverse, readxl, ipumsr]

[1] Michael Tonry, “Racial Disproportion in US Prisons.” *British Journal of Criminology* 34 (1994): 97.

[2] Marc Mauer, “The Causes and Consequences of Prison Growth in the United States.” *Punishment & Society* 3, no. 1 (2001): 9–20. <https://doi.org/10.1177/14624740122228212>.

[3] Brenden Beck, “Local Government Spending: Policing Versus Social Services.” *Annual Review of Criminology* 8, no. Volume 8, 2025 (2025): 505–28. <https://doi.org/10.1146/annurev-criminol-111523-122639>.

[4] Sarah Flood, Miriam King, Renae Rodgers, Steven Ruggles, J. Robert Warren, Daniel Backman, Etienne Breton, Grace Cooper, Julia A. Rivera Drew, Stephanie Richards, David Van Riper, and Karl C.W. Williams. IPUMS CPS: Version 13.0 [dataset]. Minneapolis, MN: IPUMS, 2025. <https://doi.org/10.18128/D030.V13.0>



[QR:
optional]